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Subject: Programming with Java	Sem: III AY: 2026-27
Assignment No: 3	
TITLE: Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.	

Problem Statement

Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Learning Objectives

To implement method overloading and use static variables and methods in Java programs. Theory

Method overloading allows multiple methods to have the same name with different parameter lists, enabling compile-time polymorphism. The static keyword is used for variables and methods that belong to the class instead of individual objects. Static members share common data among all objects and reduce memory usage. These concepts improve code flexibility, readability, and efficient resource management.

Exercise :

1. Develop a Calculator program using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.

Source code:

```
J Calculator.java
1  public class Calculator {
2
3      static int count = 0;
4
5      Calculator() // constructor
6      {
7          count++;
8      }
9
10     int add(int a, int b) {
11         return a + b;
12     }
13
14     double add(double a, double b) {
15         return a + b;
16     }
17
18     int add(int a, int b, int c) {
19         return a + b + c;
20     }
21
22     static void displayCount() {
23         System.out.println("Objects created: " + count);
24     }
25
26     public static void main(String[] args) {
27
28         Calculator obj1 = new Calculator();
29         Calculator obj2 = new Calculator();
30         Calculator obj3 = new Calculator();
31
32         System.out.println("Sum of 2 integers: " + obj1.add(10, 20));
33         System.out.println("Sum of 2 doubles: " + obj2.add(10.5, 20.5));
34         System.out.println("Sum of 3 integers: " + obj3.add(10, 20, 30));
35
36         Calculator.displayCount();
37     }
38 }
```

OUTPUT

```
PS C:\Users\sitap\OneDrive\Desktop\java\colleage ass> c::; cd 'c:\Users\sitap\OneDrive\Desktop\java\colleage ass' & java -26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' -jar 'c:\Users\sitap\OneDrive\Desktop\java\colleage ass_d78f1397\bin\colleage ass.jar'
Sum of 2 integers: 30
Sum of 2 doubles: 31.0
Sum of 3 integers: 60
Objects created: 3
PS C:\Users\sitap\OneDrive\Desktop\java\colleage ass> date

28 July 2026 20:02:59

PS C:\Users\sitap\OneDrive\Desktop\java\colleage ass> 
```

2. Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

Source code:

```

J RestaurantBilling.java > RestaurantBilling
1  public class RestaurantBilling {
2
3      static int totalOrders = 0;
4
5      // Constructor
6      RestaurantBilling() {
7          totalOrders++;
8      }
9
10     // Dine-in Bill (Food + Drinks)
11     double bill(double foodAmount, double drinksAmount) {
12         return foodAmount + drinksAmount;
13     }
14
15     // Takeaway Bill (Food + Packing Charge)
16     double bill(double foodAmount, double packingCharge, int type) {
17         return foodAmount + packingCharge;
18     }
19
20     // Delivery Bill (Food + Packing + Delivery Charge)
21     double bill(double foodAmount, double packingCharge, double deliveryCharge) {
22         return foodAmount + packingCharge + deliveryCharge;
23     }
24
25     // Static Method
26     static void displayOrders() {
27         System.out.println("Total Orders: " + totalOrders);
28     }
29
30     Run | Debug
    public static void main(String[] args) {

```

```

Run | Debug
30  public static void main(String[] args) {
31
32      RestaurantBilling order1 = new RestaurantBilling();
33      RestaurantBilling order2 = new RestaurantBilling();
34      RestaurantBilling order3 = new RestaurantBilling();
35
36      System.out.println("Dine-in Bill: Rs. " + order1.bill(foodAmount: 500, drinksAmount: 100));
37      System.out.println("Takeaway Bill: Rs. " + order2.bill(foodAmount: 400, packingCharge: 20, type: 1));
38      System.out.println("Delivery Bill: Rs. " + order3.bill(foodAmount: 350, packingCharge: 20, type: 50));
39
40      RestaurantBilling.displayOrders();
41  }
42  }

```

Output:

```
leage ass_d78f1397\bin' 'RestaurantBilling'  
Dine-in Bill: Rs. 600.0  
Takeaway Bill: Rs. 420.0  
Delivery Bill: Rs. 370.0  
Total Orders: 3  
● PS C:\Users\sitap\OneDrive\Desktop\java\colleage ass> date  
  
28 July 2026 20:06:13  
  
○ PS C:\Users\sitap\OneDrive\Desktop\java\colleage ass> █
```

Conclusion: *This experiment demonstrated the concept of **method overloading** and the use of **static variables and static methods** in Java. The Calculator application showed how the same method name can perform addition for different data types and numbers of parameters, while the Restaurant Billing application used overloaded methods to calculate bills for different order types. Static variables were used to keep track of the total number of calculations and orders, and static methods were used to display these counts. Overall, the programs improved code reusability, readability, and demonstrated important Object-Oriented Programming (OOP) concepts in Java.*